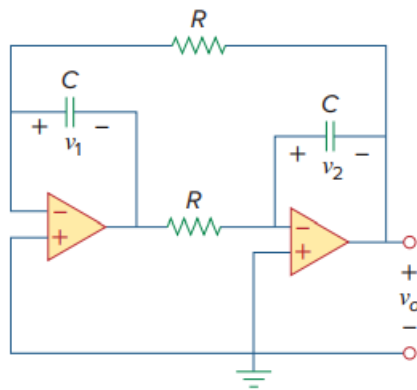


Problem

Determine the differential equation for the op amp circuit in Fig. 8.110. If $v_1(0^+) = 2\text{ V}$ and $v_2(0^+) = 0\text{ V}$, find v_o for $t > 0$. Let $R = 100\text{ k}\Omega$ and $C = 1\text{ }\mu\text{F}$.

Figure 8.110:



Step-by-step solution

Step 1 of 8

Refer to Figure 8.110 from the textbook.

Redraw the circuit as shown in figure 1.

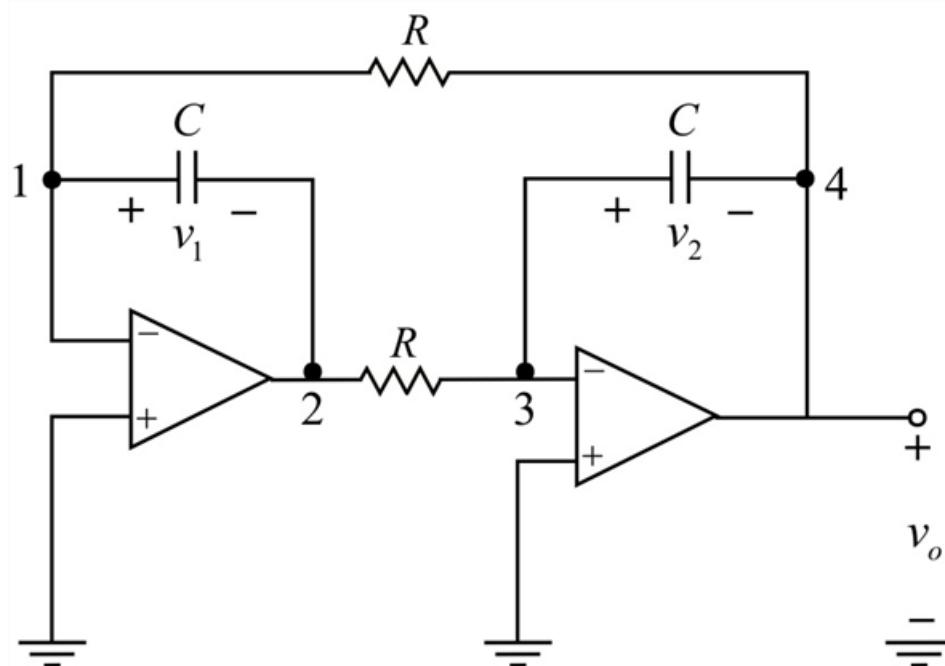


Figure 1

Step 2 of 8

Apply Nodal analysis at node 1.

$$\frac{0 - v_o}{R} + C \frac{dv_1}{dt} = 0$$

$$v_o = RC \frac{dv_1}{dt}$$

$$\frac{dv_1}{dt} = \frac{v_o}{RC} \dots\dots (1)$$

Step 3 of 8

Apply Nodal analysis at node 3.

$$\frac{0 - (-v_1)}{R} + C \frac{dv_2}{dt} = 0$$

$$C \frac{dv_2}{dt} = \frac{-v_1}{R} \dots\dots (2)$$

Differentiate with respect to t .

$$C \frac{d^2 v_2}{dt^2} = \frac{-1}{R} \left(\frac{dv_1}{dt} \right)$$

Consider equation (1).

Substitute $\left(\frac{v_o}{RC} \right)$ for $\frac{dv_1}{dt}$.

$$C \frac{d^2 v_2}{dt^2} = \frac{-1}{R} \left(\frac{v_o}{RC} \right)$$

$$\frac{d^2 v_2}{dt^2} + \left(\frac{1}{R^2 C^2} \right) v_o = 0 \dots\dots (3)$$

Step 4 of 8

Assume that the voltage at node-3 as V_3 .

The voltage at node-3 is,

$$V_3 = 0 \text{ V}.$$

Therefore, the output voltage is

$$v_o = -v_2$$

$$v_2 = -v_o$$

Step 5 of 8

Recall equation (2).

$$\frac{d^2 v_2}{dt^2} + \left(\frac{1}{R^2 C^2} \right) v_o = 0$$

Substitute $-v_o$ for v_2 .

$$\frac{d^2 (-v_o)}{dt^2} + \left(\frac{1}{R^2 C^2} \right) v_o = 0$$

$$\frac{d^2 (v_o)}{dt^2} - \left(\frac{1}{R^2 C^2} \right) v_o = 0$$

The characteristic equation is,

$$s^2 - \left(\frac{1}{RC} \right)^2 = 0$$

Substitute 100×10^3 for R , and 10^{-6} for C .

$$s^2 - \left(\frac{1}{100 \times 10^3 \times 10^{-6}} \right)^2 = 0$$

$$s^2 - 100 = 0$$

$$(s + 10)(s - 10) = 0$$

Hence, the roots are,

$$s_1 = -10, \text{ and}$$

$$s_2 = 10.$$

Step 6 of 8

The expression for v_o for $t > 0$ is,

$$v_o = Ae^{-s_1 t} + Be^{-s_2 t}$$

Substitute s_1 for s_1 , and 10 for s_2 .

$$v_o = Ae^{10t} + Be^{-10t} \dots\dots (4)$$

At $t = 0^+$,

$$\begin{aligned} v_o(0^+) &= Ae^{10 \times 0} + Be^{-10 \times 0} \\ &= A + B \end{aligned}$$

Substitute $-v_2(0^+)$ for $v_o(0^+)$.

$$-v_2(0^+) = A + B$$

Substitute 0 for $v_2(0^+)$.

$$0 = A + B$$

$$A = -B \dots\dots (5)$$

Step 7 of 8

Recall equation (2).

$$C \frac{dv_2}{dt} = \frac{-v_1}{R}$$

Replace $-v_o$ for v_2 .

$$-C \frac{dv_o}{dt} = \frac{-v_1}{R}$$

Substitute $(Ae^{10t} + Be^{-10t})$ for v_o .

$$-C \frac{d}{dt}(Ae^{10t} + Be^{-10t}) = \frac{-v_1}{R}$$

$$C[10Ae^{10t} - 10Be^{-10t}] = \frac{v_1}{R}$$

At $t = 0^+$,

$$C[10Ae^{10 \times 0} - 10Be^{-10 \times 0}] = \frac{v_1(0^+)}{R}$$

$$10A - 10B = \frac{v_1(0^+)}{RC}$$

Substitute 2 for $v_1(0^+)$, 100×10^3 for R , and 10^{-6} for C .

$$10A - 10B = \frac{2}{100 \times 10^3 \times 10^{-6}}$$

$$10A - 10B = 20$$

$$A - B = 2$$

Consider equation (5).

Replace $\diamond\diamond$ by A .

$$A + A = 2$$

$$A = 1$$

Recall equation (5).

$$A = -B$$

Substitute 1 for A .

$$B = -1$$

Step 8 of 8

Recall equation (4).

$$v_o = Ae^{10t} + Be^{-10t}$$

Substitute 1 for A , and $\diamond\diamond$ for B .

$$v_o = (e^{10t} - e^{-10t}) \text{ V}$$

Therefore, the voltage v_o for $t > 0$ is,

$$\boxed{v_o = (e^{10t} - e^{-10t}) \text{ V}}.$$